Dashboard Analysis of Canadian Community Health Survey

A Project report

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology
(Computer Science and Engineering)

Submitted to

LOVELY PROFESSIONAL UNIVERSITY

PHAGWARA, PUNJAB

SUBMITTED BY

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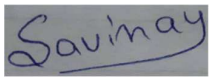
Signature of the student: 

Course Code: INT374

Under the guidance of

Ms. Sadaf Quereshi

Assistant Professor

LinkedIn URL -

https://www.linkedin.com/posts/savinaysingh_powerbi-dataanalytics-publichealth-activity-7403805277285683202-hWAG?utm_source=share&utm_medium=member_desktop&rcm=ACoAAENycI8BN9GDyDug2K-2K9jxWG5Tc0ltyo8

Preview - https://app.powerbi.com/view?r=eyJrIjoiNTI5NjI3OTMtMmE3Ny00MGQ2LTkyYzMtOGNjMzgzNDZiNTY1IiwidCI6ImUxNGU3M2ViLTUyNTEtNDM4OC04ZDY3LThmOWYyZTJkNWE0NiIsImMiOjEwfQ%3D%3D

Drive link –

https://mega.nz/folder/83I3gJgA#2EVkcqkxJAU1Pl4lzsI9yg

Dataset Link –

https://www.canada.ca/en/health-canada/services/food-nutrition/food-nutrition-surveillance/health-nutrition-surveys/canadian-community-health-survey-cchs.html

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6d •

📊 Canadian Community Health Survey — Power BI Analytics Dashboard

I've completed an end-to-end health analytics project using data from the Canadian Community Health Survey (108,000+ respondents), exploring how lifestyle, mental health, chronic conditions, and socioeconomic factors shape population health outcomes.

🔍 Key insights uncovered: • Strong income gradients in self-rated health and chronic disease burden • Clear associations between food insecurity and mental health outcomes • Rising multimorbidity with age, particularly in lower-income populations • Compounding effects of lifestyle behaviors on long-term health • Persistent inequities in health insurance access and disease burden

🛠️ Technical implementation: • Extensive data cleaning and survey code decoding in Power Query • Custom DAX measures and derived health indicators • Multi-page dashboard with cross-filtering and dynamic slicing • Focus on analytical storytelling over decorative visuals

📌 Core takeaway: Socioeconomic factors aren't just background variables—they're fundamental drivers of health inequity, with measurable impacts on mental health, chronic disease, and overall well-being.

This project strengthened my ability to translate complex public health data into actionable insights. Open to feedback and connecting with others working at the intersection of data and health!

#PowerBI #DataAnalytics #PublicHealth #HealthInformatics #DataVisualization #H

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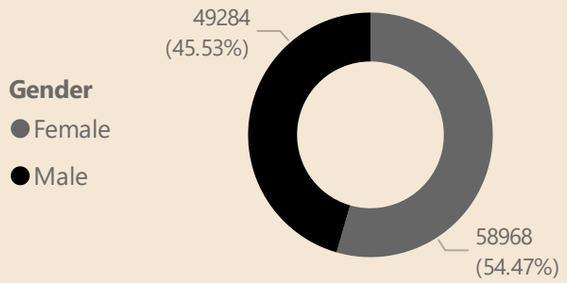
CANADIAN COMMUNITY HEALTH SURVEY OVERVIEW

Self-reported health, mental well-being, and demographics



Total Respondents

108252

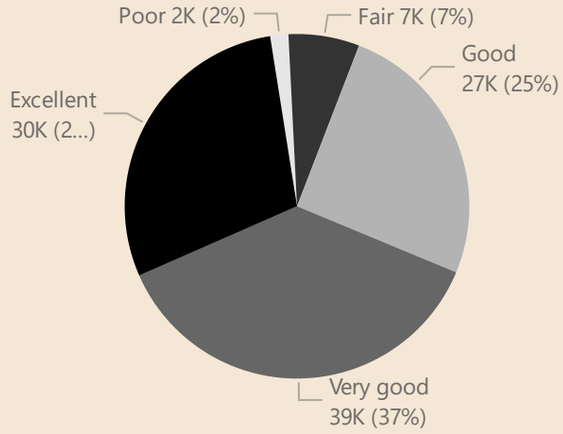
Province

All

Self-Rated General Health Status

Respondents by Age Group

Self-Rated Mental Health Status



% Good or Better Health

84.93%

Average Life Satisfaction

8.06

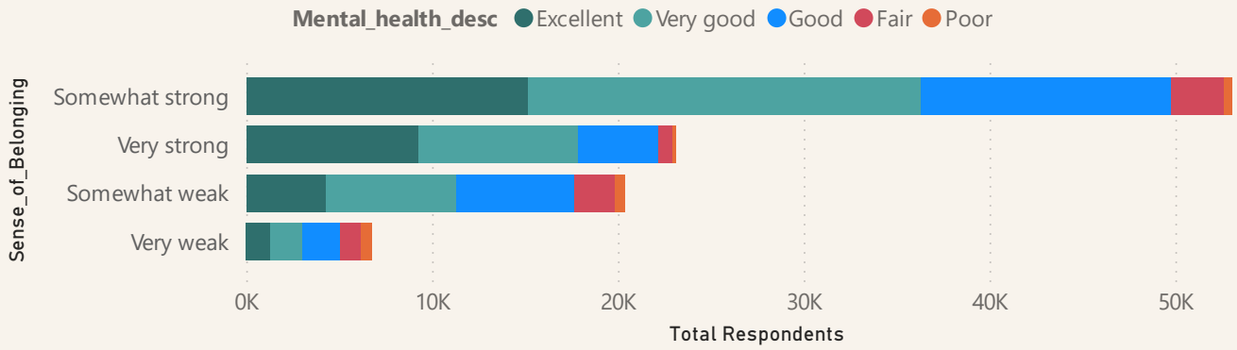
Mental Health, Stress & Well-Being

Relationship between perceived stress, mental health, work stress, and community belonging

Gender	Working_Status	Age_Group	Income Group
All	All	All	All

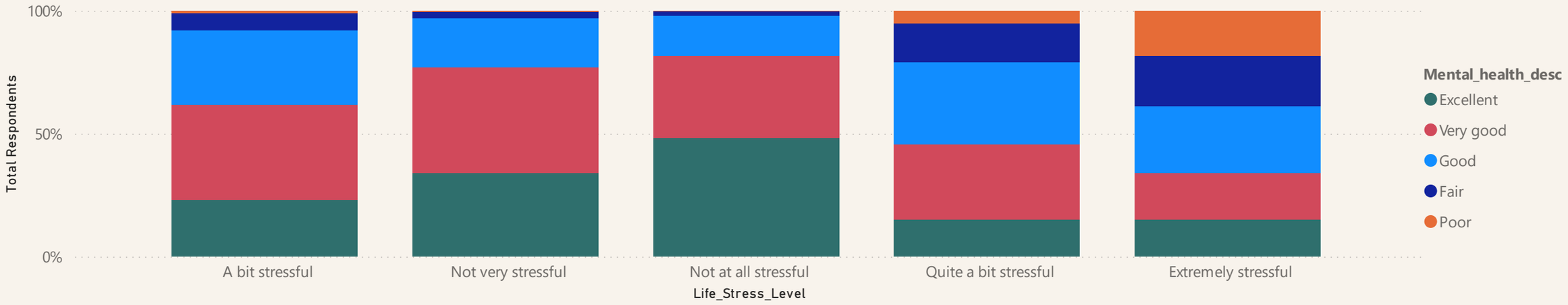
% Reporting High Stress	Average Mental Health	Average Life Satisfaction
17.4%	2.15	8.06

Mental Health by Sense of Community Belonging



Life Satisfaction by Work Stress Level

Mental Health Status by Perceived Stress Level



LIFESTYLE & HEALTH BEHAVIOURS

Physical activity, diet, and substance use patterns

7.25%
% MEETING ACTIVITY GUIDELINES

85.28
Avg Weekly Physical Activity (min)

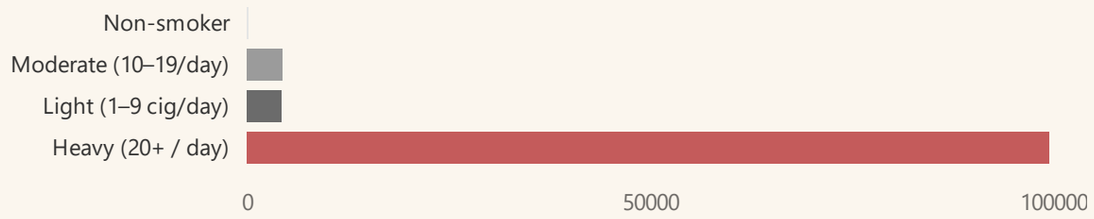
GENDER
All

AGE GROUP
All

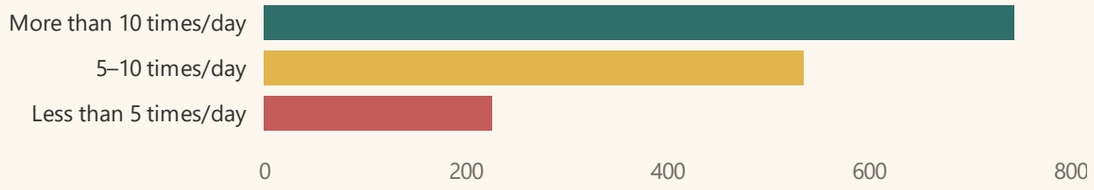
INCOME GROUP
All

WORKING STATUS
All

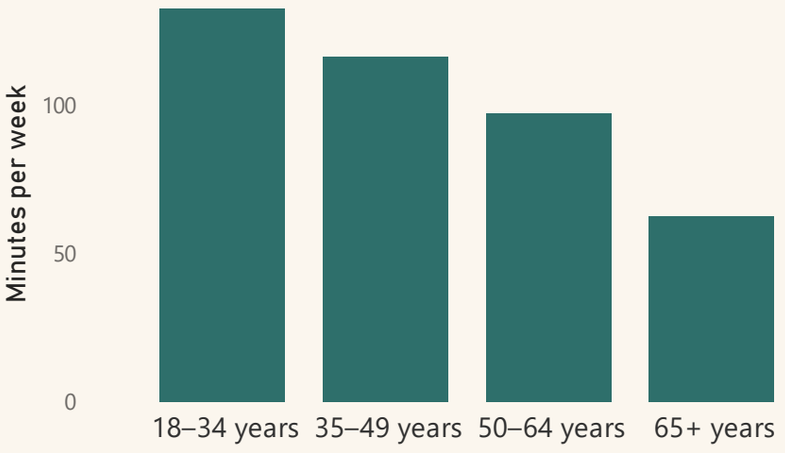
Smoking Intensity Among Smokers



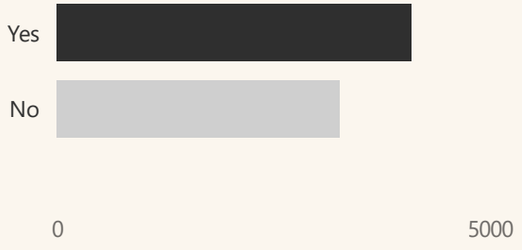
Fruit & Vegetable Intake



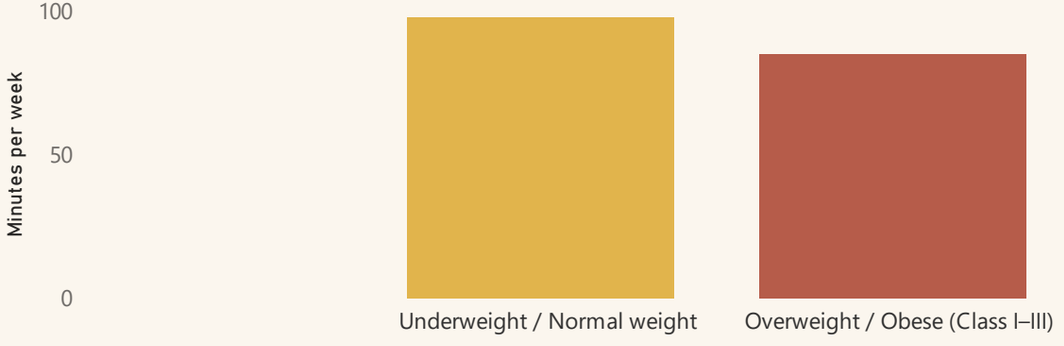
Average Weekly Physical Activity by Age Group



Alcohol Consumption in the Last 12 Months



Average Weekly Physical Activity by BMI



CHRONIC CONDITIONS & DISEASE BURDEN

Prevalence, multimorbidity, and chronic disease clustering

PROVINCE
All

GENDER
All

AGE GROUP
All

INCOME GROUP
All

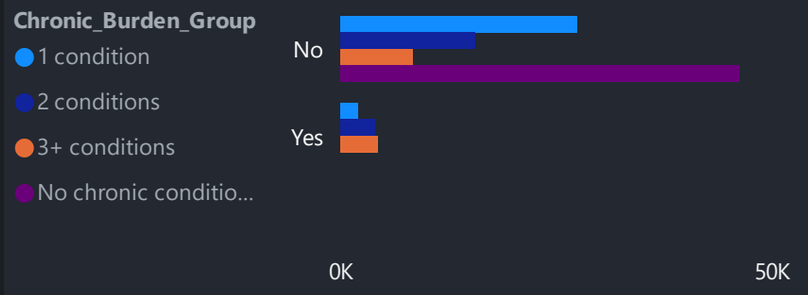
WORKING STATUS
All

1.03
Avg Chronic Burden

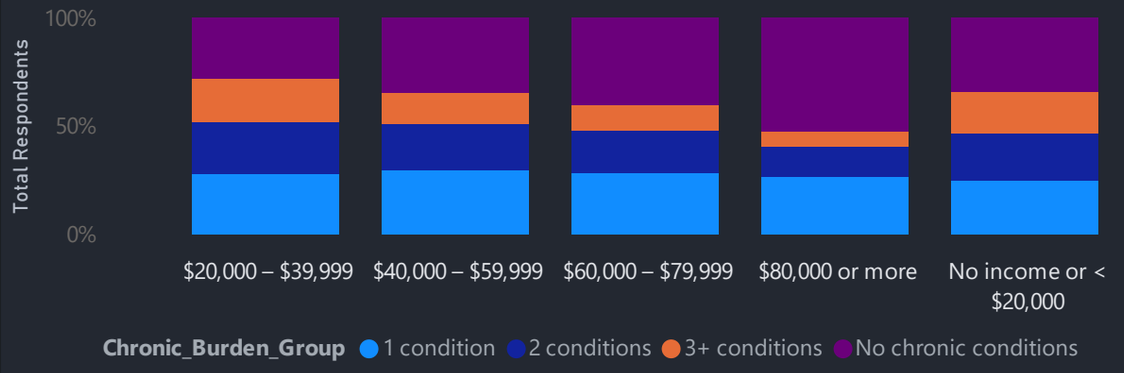
29.9%
% With Multimorbidity

57.1%
Population with ≥1 Chronic Condition

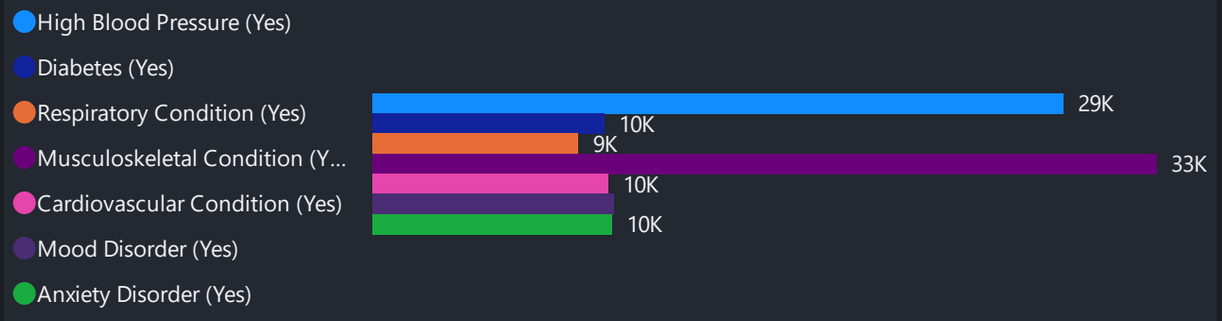
Mental–Physical Health Comorbidity Patterns



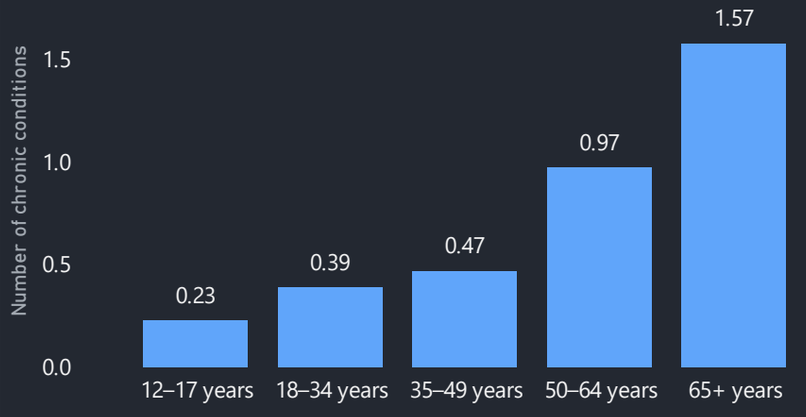
Distribution of Chronic Disease Burden by Household Income



Prevalence of Chronic Conditions



Average Number of Chronic Conditions by Age Group



SOCIOECONOMIC INEQUALITY & HEALTH OUTCOMES

Income, food security, insurance access, and population health

PROVINCE
All

GENDER
All

AGE GROUP
All

INCOME GROUP
All

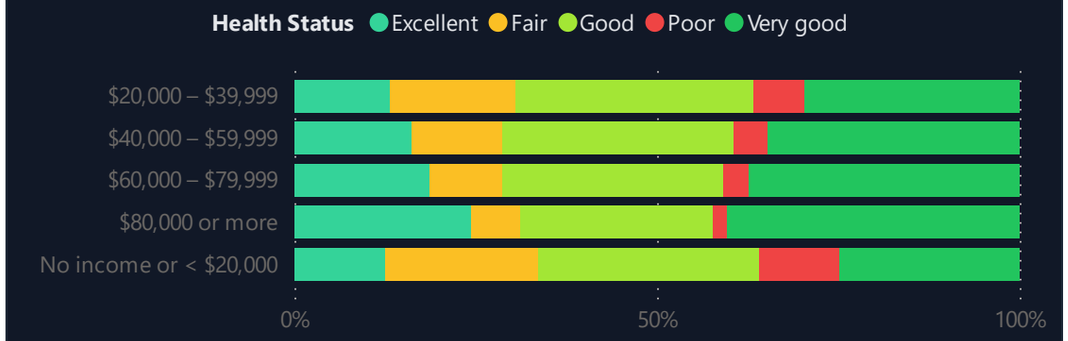
WORKING STATUS
All

Population Experiencing Food Insecurity
8.81%

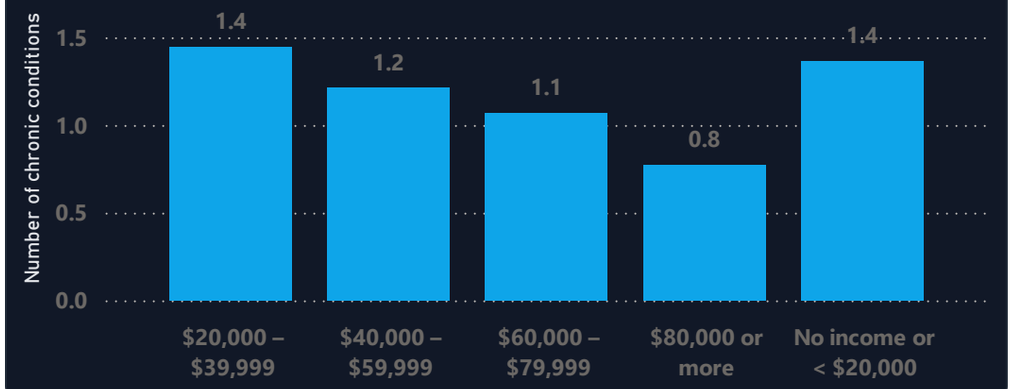
AVG CHRONIC CONDITION BURDEN
1.03

Population Lacking Health Insurance
4.20%

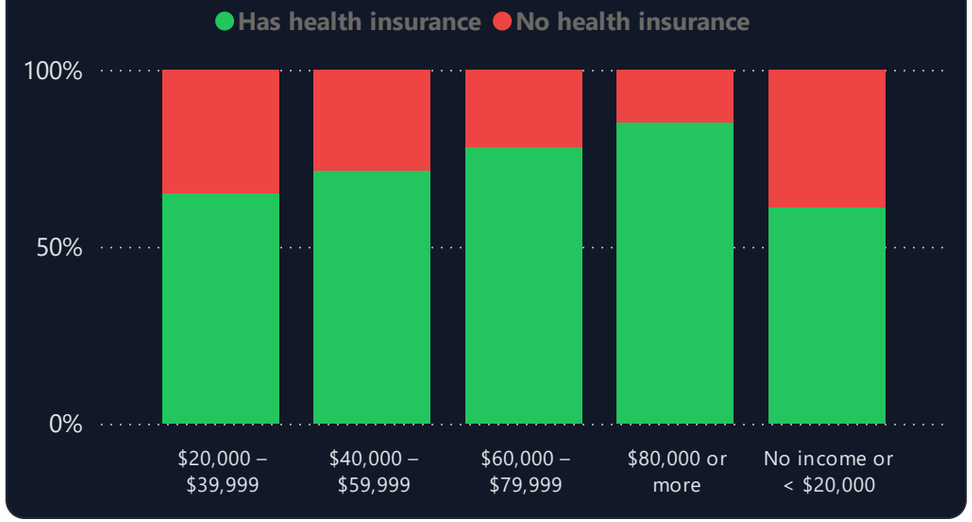
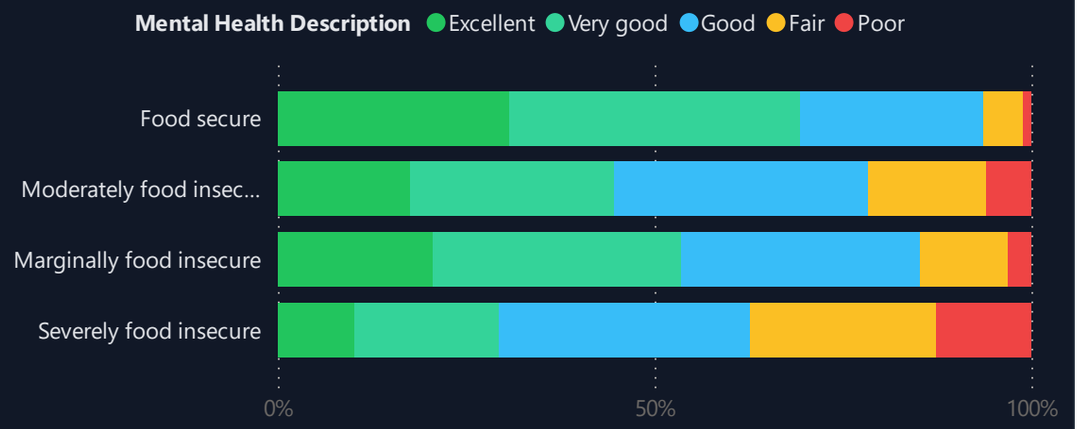
Self-Rated Health by Household Income



Average Chronic Disease Burden by Income Level



Health Insurance Coverage Across Income Groups



Mental Health Status by Food Security Level